

# Lecture 17: Nature's Thermostat (Reactivity Feedback and The Cold Water Accident)

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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## Reading Assignment

Lamarsh & Baratta (4th Edition):

- **Section 7.4** Temperature Effects on Reactivity (Doppler, Moderator/Void Coefficients).

## The Missing Piece of Kinetics

In the last two lectures, we treated Reactivity ( $\rho$ ) as an input controlled solely by the operator (rod position).

$$\rho_{total} = \rho_{rods}$$

If this were true, a reactor would be incredibly unstable. Any small rise in power would heat the core, but if  $\rho$  didn't change, the power would just keep rising forever (or until the operator intervened).

In reality, the physics of the core changes with temperature.

$$\rho_{total} = \rho_{rods} + \rho_{feedback}(T)$$

This feedback is what allows a reactor to run steadily at high power.

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## 1 The Feedback Coefficient ( $\alpha$ )

We define the reactivity coefficient  $\alpha$  as the change in reactivity per unit change in temperature:

$$\alpha = \frac{d\rho}{dT} \tag{1}$$

For stability, we require **Negative Feedback** ( $\alpha < 0$ ).

- **Scenario:** Power Rises  $\rightarrow$  Temp Rises  $\rightarrow$  Reactivity Drops ( $\rho < 0$ )  $\rightarrow$  Power Turns ( $\frac{dP}{dt} < 0$ ).
- **Result:** The system self-stabilizes.

We split this into two main components:

1. **Fuel Temperature Coefficient ( $\alpha_{fuel}$ ):** Fast response (Doppler).
2. **Moderator Temperature Coefficient ( $\alpha_{mod}$ ):** Slower response (Density).

## 2 The Fuel Temperature Coefficient (The Doppler Effect)

The fuel temperature coefficient is the single most important intrinsic safety feature of a nuclear reactor. It acts almost instantaneously, because the heat is generated *inside the fuel itself*, without relying on heat transfer to the moderator or coolant.

An increase in fuel temperature always introduces *negative reactivity*, stabilizing the reactor.

### 2.1 The Physics: Resonance Absorption in U-238

Recall the resonance capture region of U-238 (Lecture 4), often illustrated by the “valley of death” in the  $(n, \gamma)$  cross section. In the epithermal energy range (roughly a few eV to tens of keV), the capture cross section of U-238 is dominated by sharp resonance peaks.

The most important of these is the *first large resonance* at approximately  $E \approx 6\text{--}8$  eV.

- Neutrons slow down from fission energies through this region.
- The neutron flux per unit lethargy is large at low epithermal energies.
- As a result, a significant fraction of neutrons “see” the lowest-energy resonances.

These resonances are therefore responsible for most resonance absorption in U-238.

### 2.2 Doppler Broadening of Resonances

The Doppler effect does *not* primarily shift resonance energies; instead, it *broadens* them.

- **Cold Fuel:** U-238 nuclei have relatively small thermal velocities. Resonance peaks are narrow and tall. Neutrons slightly off the resonance energy may pass through without being absorbed.
- **Hot Fuel:** U-238 nuclei move rapidly due to thermal motion. From the neutron’s frame of reference, the relative energy of the interaction varies due to target motion.

The result is **Doppler broadening**:

- Resonance peaks become wider
- Peak heights decrease
- The total area under each resonance remains approximately constant

However, the broader “wings” of the resonance now absorb neutrons that previously escaped.

### 2.3 Why the Lowest Resonances Dominate

Although all resonances Doppler broaden, the *reactivity effect* is dominated by the lowest-energy resonances.

- Doppler width increases with resonance energy, but
- Neutron population decreases rapidly with energy

In a thermal reactor, the epithermal flux behaves approximately as

$$\phi(E) \propto \frac{1}{E}$$

As a result:

- There are many more neutrons near 8 eV than near 800 eV
- Doppler broadening of the first few U-238 resonances captures far more neutrons than broadening of higher-energy resonances

Consequently:

- The first resonance contributes the majority of the Doppler effect
- Higher-energy resonances provide progressively smaller contributions

## 2.4 Net Reactivity Effect

As fuel temperature increases:

- Resonance absorption in U-238 increases
- More neutrons are removed before reaching thermal energies
- Fewer neutrons cause fission in U-235

$$\text{Fuel Temperature } \uparrow \Rightarrow \text{U-238 Resonance Absorption } \uparrow \Rightarrow \text{Reactivity } \downarrow$$

This effect is called the **Doppler coefficient** and is always negative.

## 2.5 Safety Implications

The Doppler effect acts on a *nuclear timescale*.

If a reactor experiences a rapid reactivity insertion:

- Power increases immediately
- Fuel temperature rises within milliseconds
- Doppler broadening introduces prompt negative reactivity

This feedback rapidly terminates prompt excursions before fuel melting or mechanical disassembly can occur.

The Doppler effect is a fundamental reason why commercial reactors cannot undergo a nuclear detonation: the fuel itself provides an automatic, instantaneous stabilizing response.

## 3 The Moderator Temperature Coefficient

This effect is delayed because heat must transfer from the fuel cladding to the coolant (water).

### 3.1 The Physics: Density Changes

In a Light Water Reactor (PWR/BWR), water serves two roles:

1. **Moderator:** Slows neutrons down (increases  $k$ ).
2. **Absorber:** Captures neutrons (decreases  $k$ ).

As water heats up, it expands (density  $\rho_w$  decreases).

- Fewer H atoms/cm<sup>3</sup>  $\implies$  Less Moderation.
- Neutrons stay fast longer  $\implies$  More leakage and more resonance capture in U-238.
- **Result:**  $k_{eff}$  decreases.

$$\alpha_{mod} < 0 \quad (\text{Under-moderated Design})$$

**Design Note:** We intentionally design PWRs to be "Under-moderated" (less water than optimal). If we lose water (density drops), the moderation gets worse, and the reactor shuts down. If we designed it "Over-moderated," a loss of density would move us *closer* to the optimum, increasing power (Positive Feedback)—a characteristic of the Chernobyl design (RBMK).

## 4 The Power Defect

When bringing a reactor from Hot Zero Power (HZP) to Full Power (FP), the fuel temperature rises significantly (e.g., from 300°C to 1000°C). Because of the negative coefficients ( $\alpha_{fuel} + \alpha_{mod}$ ), this temperature rise inserts a large amount of negative reactivity.

This is called the **Power Defect**.

$$\text{Power Defect} = \int_{T_{HZP}}^{T_{FP}} \alpha(T) dT$$

To reach full power, the operator must withdraw control rods not just to make it critical, but to **compensate** for this defect. You have to "pay" reactivity to buy power. This is also why you have to initially bank extra  $k > 1$  reactivity into the reactor, compensated for by poisons or control rods that are removed over the life of the core.

## 5 Stability Analysis: The "Load Follow" Capability

This negative feedback allows the reactor to follow the steam turbine load without operator action.

1. **Steam Demand Increases:** Turbine valves open.
2. **Coolant Cooled:** More heat is drawn from the steam generator. The water returning to the core is colder.
3. **Reactivity Added:** Colder water  $\rightarrow$  Higher Density  $\rightarrow$  Better Moderation  $\rightarrow \rho > 0$ .
4. **Power Rises:** The reactor power rises to meet the new heat demand.
5. **Equilibrium:** The fuel heats up until the Doppler effect ( $\rho < 0$ ) exactly balances the colder water ( $\rho > 0$ ).

## 6 Case Study: The Cold Water Accident

While negative feedback ( $\alpha < 0$ ) is essential for stability against *overheating*, it creates a vulnerability to *overcooling*. This is a classic example of a "Double-Edged Sword" in engineering.

### 6.1 The Scenario

This scenario is most severe at low power, when Doppler feedback is weakest and moderator temperature changes dominate. Consider a naval reactor or a multi-loop commercial PWR operating at low power. Suppose one of the coolant loops has been idle (pump off). The water in that loop cools down to 150°C, while the core is at 300°C. If the operator suddenly turns the pump on, a "slug" of cold water enters the core. Because this insertion occurs faster than precursor decay, point kinetics reverts to prompt behavior.

### 6.2 The Physics of the Accident

Because  $\alpha_{mod}$  is negative, a drop in temperature causes a rise in reactivity:

$$\Delta\rho = \alpha_{mod} \times \Delta T$$

$$\Delta\rho = (\text{Negative}) \times (\text{Negative}) = \textbf{Positive Reactivity}$$

If the slug is cold enough and enters fast enough, the reactivity insertion can exceed the delayed neutron fraction ( $\beta$ ).

$$\rho > \beta \implies \textbf{Super-Prompt Critical}$$

The reactor power could rise on a millisecond period, potentially leading to a steam explosion (similar to the [SL-1 accident](#), although that was caused by rapid removal of the core rods rather than coolant temperature drop - same effect, though) before control rods could mechanically drop.

### 6.3 Why hasn't it happened here?

This is a known **\*\*Design Basis Accident (DBA)\*\***. Engineers prevent it through:

- **Interlocks:** Control systems physically prevent a main coolant pump from starting if the temperature difference ( $\Delta T$ ) between the loop and the core is too high.
- **Procedures:** Strict administrative controls **must** prevent the startup of idle loops while critical.

**The Lesson:** We design the physics for stability (Negative  $\alpha$ ), but we must design the controls to protect against the consequences of that physics (Cold Water Interlocks) and, as much as possible, operator error.

## 7 The Extreme Limit: Negative Void Coefficient

Up to this point, we have discussed the moderator temperature coefficient in the regime where the coolant remains liquid and its density changes smoothly with temperature. The logical extreme of this effect occurs when the moderator does not merely expand, but is **physically removed** from the core by boiling or void formation.

This is described by the **Void Coefficient of Reactivity**:

$$\alpha_{void} = \frac{d\rho}{dV_{void}} \quad (2)$$

where  $V_{void}$  is the fraction of coolant volume replaced by steam (or gas).

## 7.1 Light Water Reactors: Loss of Moderator = Loss of Reactivity

In a Light Water Reactor (PWR or BWR), water is the *only* effective moderator. If it boils:

- Hydrogen density collapses.
- Neutrons are no longer efficiently thermalized.
- The spectrum hardens dramatically.
- Resonance absorption in U-238 increases and leakage rises.

The net effect is a large, strongly negative reactivity insertion:

$$\alpha_{void} < 0 \quad (\text{LWRs}) \quad (3)$$

**Physical Interpretation:** If the reactor overheats badly enough to boil its coolant, it shuts itself down even harder. The chain reaction cannot be sustained without liquid water.

This is not a subtle effect. In a BWR, the void coefficient dominates normal power control:

- More power  $\rightarrow$  more boiling
- More boiling  $\rightarrow$  less reactivity
- Power stabilizes automatically

**Key Safety Insight:** In an LWR, a Loss-of-Coolant Accident (LOCA) is simultaneously a *Loss-of-Moderator Accident*. The nuclear chain reaction dies long before fuel damage becomes the limiting concern. Residual decay heat from fission products still must be removed to avoid a "slow motion meltdown" such as occurred in Fukushima, however.

## 7.2 Why This Is the Opposite of the Cold Water Accident

The Cold Water Accident and Void Formation represent two ends of the same feedback mechanism:

| Event           | Moderator Effect  | Reactivity               |
|-----------------|-------------------|--------------------------|
| Cold Water Slug | Density increases | $\rho > 0$ (dangerous)   |
| Normal Heating  | Density decreases | $\rho < 0$ (stabilizing) |
| Boiling / Voids | Moderator removed | $\rho \ll 0$ (shutdown)  |

The Cold Water Accident is dangerous precisely because it moves the system in the *wrong direction along an otherwise stabilizing curve*. Void formation moves it violently in the *right direction*.

### 7.3 A Crucial Historical Contrast: RBMK Reactors

Not all reactors share this behavior.

The RBMK design used at Chernobyl employed:

- Graphite as the moderator (solid, cannot boil)
- Light water primarily as a coolant and absorber

When water boiled in an RBMK:

- Moderation remained (graphite)
- Neutron absorption by water decreased

The result was:

$$\alpha_{void} > 0 \quad (\text{RBMK}) \quad (4)$$

This meant that boiling *added* reactivity instead of removing it — a fundamentally unstable feedback loop that played a central role in the Chernobyl accident.

### 7.4 The Big Picture

Negative temperature and void coefficients form a layered safety hierarchy:

1. **Fuel Doppler:** milliseconds
2. **Moderator heating:** seconds
3. **Void formation:** ultimate shutdown mechanism

**Final Takeaway:** In modern Light Water Reactors, overheating cannot lead to runaway power. If temperature feedback fails, boiling removes the moderator itself — and the chain reaction dies.

## 8 Advanced Concepts: Reactivity Feedback in Fast Reactors

All the feedback mechanisms we have discussed so far (Doppler in the 1-1000 eV range, Moderator Density) rely on thermal or slowing-down neutrons. But what happens in a Fast Reactor (like a Sodium-Cooled Fast Reactor), where there is no moderator and the average neutron energy is  $> 100$  keV?

Furthermore, as we saw in Lecture 15, if the reactor runs on Plutonium-239, the delayed neutron fraction is tiny ( $\beta \approx 0.0021$ ). The margin to prompt critical is incredibly narrow. How do we keep the reactor stable?

### 8.1 The Weakened Doppler and the Void Coefficient

- **Doppler:** Fast reactors still have Doppler broadening, but because the neutron spectrum is so fast, very few neutrons interact in the resonance region. The Doppler coefficient is negative, but much weaker than in a Light Water Reactor.
- **Coolant Density (Sodium Void):** Liquid sodium is not a moderator, but it does slightly scatter and soften the neutron spectrum. If the sodium heats up and expands (or boils), the spectrum "hardens" (gets faster). For Pu-239, a faster spectrum actually increases the number of neutrons produced per fission ( $\eta$ ). Therefore, in large fast reactors, the coolant density coefficient can actually be **positive**.

## 8.2 The Savior: Thermal-Mechanical Expansion

To overcome the small  $\beta$  and the potentially positive coolant coefficient, fast reactors are designed to leverage the physical thermal expansion of the metals in the core to increase **neutron leakage** ( $P_{NL}$ ).

1. **Axial Fuel Expansion (Prompt):** As the power spikes, the metallic fuel pins heat up instantly and physically elongate. This effectively decreases the density of the fuel in the active core region and allows neutrons to leak out the top and bottom. This is the fast reactor's equivalent of the instant Doppler save.
2. **Radial Core Expansion (Delayed):** As the coolant heats up, the metal grid plates supporting the core expand. The fuel assemblies spread apart (like a blooming flower), massively increasing radial neutron leakage.
3. **Control Rod Drive Line Expansion (Delayed):** The control rods are suspended from the top of the reactor tank by long metal drive lines. As the hot coolant pool heats these lines, the metal elongates, passively pushing the control rods deeper into the core.

**Case Study: EBR-II Passive Safety Tests (1986)** To prove these feedback mechanisms worked, engineers at the Experimental Breeder Reactor II (EBR-II) in Idaho ran the reactor at 100% power and intentionally tripped the primary cooling pumps while simultaneously disabling the SCRAM system (Loss of Flow Without Scram - LOFWS).

Because of the massive negative reactivity inserted by the radial core expansion and control rod drive line expansion, the reactor smoothly shut itself down to a safe decay-heat level without operator intervention and without melting a single fuel pin.

- **Planchon, H. P., et al. (1987):** *Implications of the EBR-II Inherent Safety Demonstration Test*. Nuclear Engineering and Design, 101(1), 75-90. (The foundational paper detailing the historic April 3, 1986, Loss-of-Flow and Loss-of-Heat-Sink tests).
- **Argonne National Laboratory (2022):** *SAS4A/SASSYS-1 Validation with the EBR-II Tests SHRT-17 and SHRT-45R* (ANL/NSE-22/16). (A DOE report containing plots of the actual measured power and temperature transient curves demonstrating the passive shutdown during the unprotected SHRT-45R test).

## 9 Summary

- **\*\* $\alpha_{fuel}$  (Doppler):\*\*** Instant negative feedback. Prevents excursions.
- **\*\* $\alpha_{mod}$  (Density):\*\*** Delayed negative feedback. Provides load-following stability.
- **\*\*Safety Paradox:\*\*** The same negative coefficient that prevents meltdown during overheating can cause a prompt-critical excursion during overcooling (Cold Water Accident).

## 10 Case Study: TRIGA Reactors and the Ultimate "Save"

If \$1.00 is the ultimate danger limit, how do university research reactors (like the TRIGA design) routinely "pulse" their reactors to \$2.00 or even \$3.00 on purpose?



In a TRIGA reactor, the operator fires a pneumatic cylinder that ejects a control rod in a fraction of a second, intentionally making the reactor prompt critical. The power skyrockets on a millisecond period, peaking at upwards of 22,000 MW, producing a brilliant Cherenkov blue flash.

Why doesn't it melt? In Lecture 15, we discussed how U-238 Doppler broadening is a "first line of defense," but that it's often not quite big enough to stop a massive prompt excursion before the fuel melts; the reactor relies on the water (moderator) heating up to fully shut down, which takes *seconds* due to heat conduction.

The TRIGA completely bypasses this conduction delay by using **Uranium Zirconium-Hydride (U-ZrH)** fuel. The hydrogen (the moderator) is atomically bound within the fuel matrix itself, acting as a quantum harmonic oscillator within the crystal lattice.

When the uranium fissions, it heats the bound hydrogen *instantly*. As the hydrogen lattice heats up, the atoms vibrate wildly. When thermal neutrons bounce off these hot hydrogen atoms, they gain a discrete quantum of energy (upscattering), instantly shifting the thermal neutron energy spectrum "warmer."

Because the U-235 fission cross-section approximately follows a  $1/v$  dependence in this thermal region, these "warmed" neutrons suddenly have a drastically lower probability of causing fission. This spectrum hardening alone provides a massive, instantaneous negative reactivity insertion.

Furthermore, to allow modern TRIGAs to run for decades without refueling, the fuel is deliberately loaded with extra uranium and alloyed with a burnable poison called **Erbium-167**. Erbium has a massive absorption resonance around 0.46 eV. As the hot hydrogen pushes the neutrons to higher energies to escape U-235, they are instantly swallowed by the Erbium "wall."

Together, the  $1/v$  drop-off and the Erbium resonance pull the core back well below  $\beta$ , safely shutting the pulse down in less than 30 milliseconds—long before the heat ever reaches the water coolant.

## The Aftermath: Post-Pulse Physics and Timescales

A common student question is: *What happens after the blue flash?* If the temperature feedback is what shut the reactor down, won't it just go prompt critical again as soon as the fuel cools off in the water?

The answer comes down to a race between mechanical systems and thermodynamics. Here is the exact timeline of a TRIGA pulse:

- **t = 0 seconds:** The pneumatic cylinder fires, ejecting the transient control rod.  $\rho$  jumps to  $\beta$ .
- **t = 0.015 seconds (15 ms):** Reactor power peaks at  $\approx 20,000$  MW. The fuel temperature spikes to  $\approx 400^\circ\text{C}$ .
- **t = 0.030 seconds (30 ms):** The Erbium-167 and U-ZrH temperature feedback crushes the chain reaction. Power drops by several orders of magnitude.

At this exact moment, the control rod is **still physically out of the core**. The reactor is being held subcritical entirely by its own heat. It settles into a "tail" of low, steady power driven by the decay of delayed neutron precursors, perfectly balanced by the hot fuel.

- **t = 1 to 2 seconds:** The pneumatic pressure in the transient rod cylinder is automatically vented. Gravity pulls the heavy control rod back down into the core, driving the reactivity deeply negative.

- **t = 10 to 20 seconds:** The thermal time constant of the U-ZrH fuel kicks in. Heat finally conducts across the cladding and into the pool water. You will see bubbles rise to the surface of the pool as the water immediately surrounding the elements boils.

**The Engineering Takeaway:** The shutdown physics (the negative temperature coefficient) operates in **milliseconds**, safely killing the excursion. However, the heat transfer to the coolant takes **seconds**. This massive disparity in timescales gives the mechanical safety systems (the gravity-drop of the control rod) plenty of time to re-insert the physical poison before the core can cool down and accidentally restart the chain reaction.

## Why Build a TRIGA? Applications of Pulsing

While avoiding a meltdown is a great safety feature, the ability to pulse a reactor to 22,000 MW on purpose has incredible scientific and economic value.

**1. Simulating Reactor Accidents (Transient Testing)** To test how new commercial nuclear fuels or reactor components will survive a Reactivity Initiated Accident (RIA), engineers must subject them to a massive, instantaneous spike in neutron flux. Pulsing a TRIGA reactor allows engineers to recreate the extreme conditions of a prompt critical excursion inside a safe, controlled test chamber.

**2. High-Flux Access without High-Flux Costs** A steady-state 1,000 MW commercial reactor requires massive cooling towers, gigantic pumps, and thick pressure vessels. A TRIGA can achieve the same neutron flux ( $\approx 10^{16}$  n/cm<sup>2</sup>-s) for a few milliseconds while maintaining a time-averaged power of just a few kilowatts. This provides researchers with brief bursts of world-class neutron flux using nothing more than a natural-convection swimming pool for cooling.

**3. Neutron Activation Analysis (NAA)** Researchers can place a sample of unknown material (e.g., a geological core, forensic evidence, or an archaeological artifact) into the core and pulse the reactor. The intense burst of neutrons is absorbed by the sample, transmuting trace elements into unstable radioactive isotopes. By analyzing the specific spectrum of delayed gamma rays emitted as the sample decays, scientists can identify the exact elemental composition down to the parts-per-billion level without chemically destroying the sample.

**4. Neutron Radiography** Neutron radiography is similar to an X-ray, but it uses a beam of neutrons. While X-rays bounce off dense heavy metals and pass right through organics, neutrons easily penetrate heavy metals (like lead or steel) but are heavily scattered by light elements (like the hydrogen found in water, plastics, or oil). TRIGA neutron beams are used to take internal "pictures" of aircraft engine turbine blades to check for microscopic water or oil blockages inside solid titanium casings.

### References for Further Reading:

- **General Atomics (GA):** *TRIGA Nuclear Reactors*. (GA has installed over 60 TRIGA reactors across 24 countries, primarily for universities and medical centers).
- **IAEA (2016):** *History, Development, and Future of TRIGA Research Reactors* (Technical Reports Series No. 482).

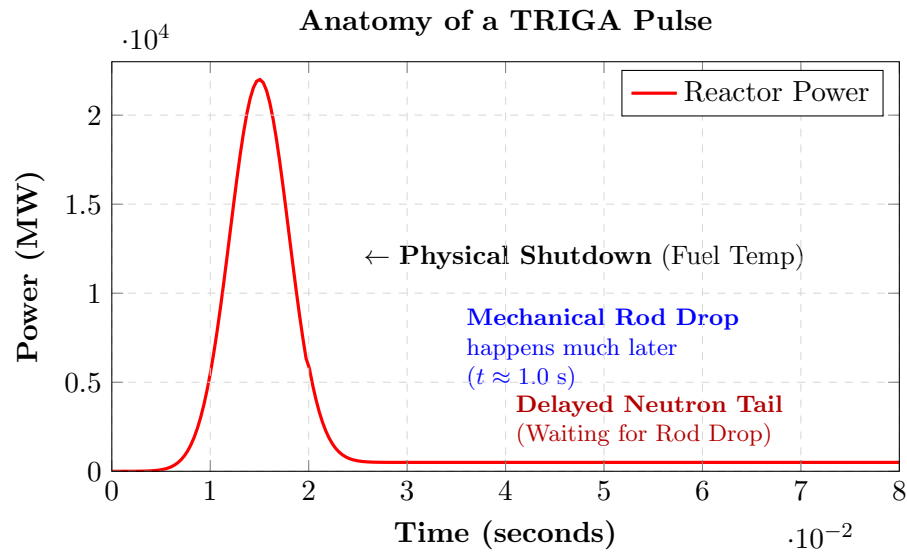


Figure 1: **The Timescale Gap:** Note the "tail" of power (approx. 0.5 - 1 MW) that persists after the pulse. This is sustained by delayed neutrons and is only terminated when the mechanical rods finally drop seconds later.